TEXORA · LECTURE SHEET

Viewing Angles, Rendering and Simulation with FEA

Presenting and proving a design: projection and view systems carried into 3D, exploded and section views, photorealistic rendering, and finite element analysis — mesh, boundary conditions, von Mises stress, factor of safety, and the hand calculation that keeps the software honest.

ME-1202 · ENGINEERING DRAWING-II · TOPIC 2 · LEVEL 1 TERM 2

1. Views — from the drawing board to the model

A 3D model is inspected through **standard views** inherited from orthographic projection (ME-1102): front, top, side, and isometric, placed on drawings by either the **first-angle** system (view placed opposite the viewing direction — the ISO/Bangladesh convention) or **third-angle** (view placed on the viewing side — common in the USA). The projection symbol on the drawing declares which; misreading it mirrors the part. Inside the CAD scene the same geometry serves more views: **section views** (cut the model on a plane to expose bores, walls and assemblies' internals), **detail views**, and the **exploded view** — components slid apart along their assembly axes with trail lines, the universal language of assembly manuals and spare-parts catalogues for textile machinery.

2. Rendering — making the model believable

Rendering turns geometry into a presentation image: assign appearances (brushed steel, anodised aluminium, fabric textures for the yarn models of Topics 3–4), place lights and an environment (HDRI in Blender; scenes in SOLIDWORKS Visualize/PhotoView 360), set a camera with perspective and depth of field, and ray-trace. Ray tracing follows light physically — reflections, refraction, soft shadows — so a render can sell a design, check colours and finishes before tooling, and expose visual flaws (interference, missing parts). Rendering proves nothing about strength: that is simulation's job.

3. FEA — the finite element idea

Finite element analysis solves stress problems that formulas cannot reach (odd shapes, holes, contact) by chopping the part into small **elements** joined at **nodes** — the **mesh**. Each element is a simple spring-like relation; assembled, they form one huge system **K u = F** (stiffness × displacement = load) which

the computer solves, recovering strains and stresses element by element. The workflow in SOLIDWORKS Simulation (and equivalently in COMSOL/SimScale-style tools):

1. **Material:** assign real properties (E, ν, σ_y). Wrong material, wrong answer.
2. **Fixtures (boundary conditions):** restrain the part as it is really held — a bolted face, a bearing journal — no more, no less.
3. **Loads:** forces, pressures, torques, gravity, centrifugal — where and as they truly act.
4. **Mesh:** finer where stress changes fast (fillets, holes); run a **convergence check** — refine until the key stress changes by only a few percent.
5. **Solve and read: von Mises stress** against yield (the ductile-failure criterion), displacement plots (exaggerated scale — check the number, not the picture), and the **factor of safety** plot, FoS = σ_y / σ_{vm}.

4. The hand check — never trust a naked contour plot

WORKED EXAMPLE — VERIFY THE FEA OF A CANTILEVER

Bracket: 200 mm long, cross-section 20 mm (high) × 10 mm (wide), fixed at one end, 100 N downward at the tip, steel (E = 200 GPa).

Bending moment at the wall: $M = PL = 100 \times 0.200 = 20$ N·m.

$I = b h^3/12 = 0.010 \times 0.020^3/12 = 6.67 \times 10^{-9}$ m⁴; c = 10 mm.

$\sigma_{max} = Mc/I = 20 \times 0.010/(6.67 \times 10^{-9}) =$ **30 MPa** at the fixed end.

Tip deflection $\delta = PL^3/(3EI) = 100 \times 0.2^3/(3 \times 200 \times 10^9 \times 6.67 \times 10^{-9}) =$ **0.20 mm**.

A competent FEA run must land close to 30 MPa and 0.20 mm. If it reports 300 MPa, suspect the setup (point-load singularity, wrong units or fixture), not the beam formulas.

Classic traps: a load applied to a single point or knife edge creates a **stress singularity** — the peak grows forever with refinement and must be ignored or spread over a real contact area; fully fixing a whole face over-stiffens bolted joints; forgetting self-weight in large parts; and reading the autoscale legend (bright red may mean 5 MPa). FEA is an instrument: calibrate it against a case you can solve by hand, every time the stakes matter.

5. Key points

- First-angle vs third-angle decides where views sit; section and exploded views are the 3D-era additions.
- Rendering = appearance + light + camera + ray tracing; it validates looks, never strength.

- FEA: mesh → K u = F → stress. The recipe is material, fixtures, loads, mesh, solve — in that order, honestly.
- Judgement tools: von Mises vs yield, FoS = σ_y/σ_vm, convergence by refinement, hand-calculation cross-check.
- Beware singularities at point loads and sharp re-entrant corners; beware over-constraint.

6. Practice

Answer in your own words; brief model answers follow.

1. A steel cantilever 150 mm long with a 15 × 15 mm square section carries 200 N at its tip. Predict, by hand, the peak bending stress and tip deflection that an FEA study should reproduce (E = 200 GPa).
2. An FEA report shows von Mises stress peaking at 120 MPa in a part whose material yields at 360 MPa. Find the factor of safety and state the verdict for a lifting component that requires FoS ≥ 3.
3. Each time the mesh is refined, the reported peak stress at one sharp internal corner rises — 180, 240, 310 MPa — while stresses elsewhere settle. Explain and state the correct engineering response.
4. A drawing shows the side view placed to the *left* of the front view. Which projection system is in use, and what happens if a machinist reads it as the other system?
5. Why is fixing an entire bolted flange face (all nodes, all directions) usually a worse boundary condition than fixing only the bolt-hole cylinders and letting the face bear on a contact surface?
6. Name three things a photorealistic render can decide for a new sewing-machine housing, and two important things it cannot.

Answer key

1. M = 200 × 0.150 = 30 N·m. I = 0.015 × 0.015³/12 = 4.22 × 10^{-9} m⁴; c = 7.5 mm. σ = Mc/I = 30 × 0.0075/(4.22 × 10^{-9}) = 53.3 MPa. δ = PL³/(3EI) = 200 × 0.150³/(3 × 200 × 10^{9} × 4.22 × 10^{-9}) = 0.675/2,531 = 2.67 × 10^{-4} m ≈ 0.27 mm. An FEA landing near 53 MPa and 0.27 mm is trustworthy; far off means the setup is wrong.
2. FoS = 360/120 = 3.0. It exactly meets the stated minimum — acceptable on paper, but with no margin for load surprises, corrosion thinning or material scatter, a prudent engineer thickens the section or improves the material before signing off a lifting part.
3. A stress singularity: at a perfectly sharp re-entrant corner the theoretical elastic stress is infinite, so finer meshes chase infinity and the “peak” is a numerical artefact. Response: read the stress a small distance away from the corner, or model the real manufactured radius (no machined corner is truly sharp) — and remember the corner is still a genuine crack-starter needing a fillet, which is the design fix.

4. First-angle projection (the view is placed opposite the side it shows). Read as third-angle, the machinist builds the mirror image — features on the wrong hand; symmetric parts survive, handed parts (brackets, covers with offset bosses) are scrapped. The projection symbol in the title block exists to prevent exactly this.
5. Full-face fixing forbids the face to lift, warp or slide at all, making the joint artificially rigid and pulling stress away from where reality puts it (bolt tension and local bearing). Fixing the bolt holes and allowing frictionless contact on the mating face reproduces the real load path — the plate bears, the bolts clamp — so stresses near holes and at the contact edge come out honest.
6. Can decide: colour and finish combinations, proportions and styling acceptance, control-panel legibility and branding placement (and assembly look via exploded renders). Cannot decide: whether the housing cracks under motor vibration (needs FEA/test) or whether the plastic survives heat and cleaning chemicals (needs material data/testing). A render is a photograph of an opinion about looks — nothing more.

Texora — course notes prepared from the official syllabus of B.Sc. in Textile Engineering · Made by Sakib